

SQL DATA ANALYSIS PROJECT

Mission SafeCity

Analyzing Women's Safety Incidents Across the City to Uncover Risk Patterns, Response Gaps, and Infrastructure Priorities

20,000

INCIDENTS ANALYZED

100

LOCATIONS COVERED

4 Yrs

2022 – 2025

25

SQL QUERIES BUILT

Objectives & Analytical Approach

Mission SafeCity examines women's safety incident records — collected via mobile apps, smartwatches, and public panic buttons — to help city safety planners understand where incidents cluster, how fast responders arrive, and which locations most urgently need infrastructure investment.



Identify Patterns

Uncover the most common incident types and when/where they occur



Flag High-Risk Zones

Rank locations and city zones by incident volume and risk level



Evaluate Response

Measure response speed across shifts, devices and panic-button use



Audit Infrastructure

Compare CCTV, streetlight and patrol coverage against incident load



Score Priorities

Build a composite score to rank locations for intervention



Apply Advanced SQL

CTEs, window functions, ranking & lag analysis across 25 queries

Dataset & Relational Schema

A star style schema built in MySQL Workbench: one central incidents fact table joined to five dimension tables, plus a flat pre-normalization staging table (safety).

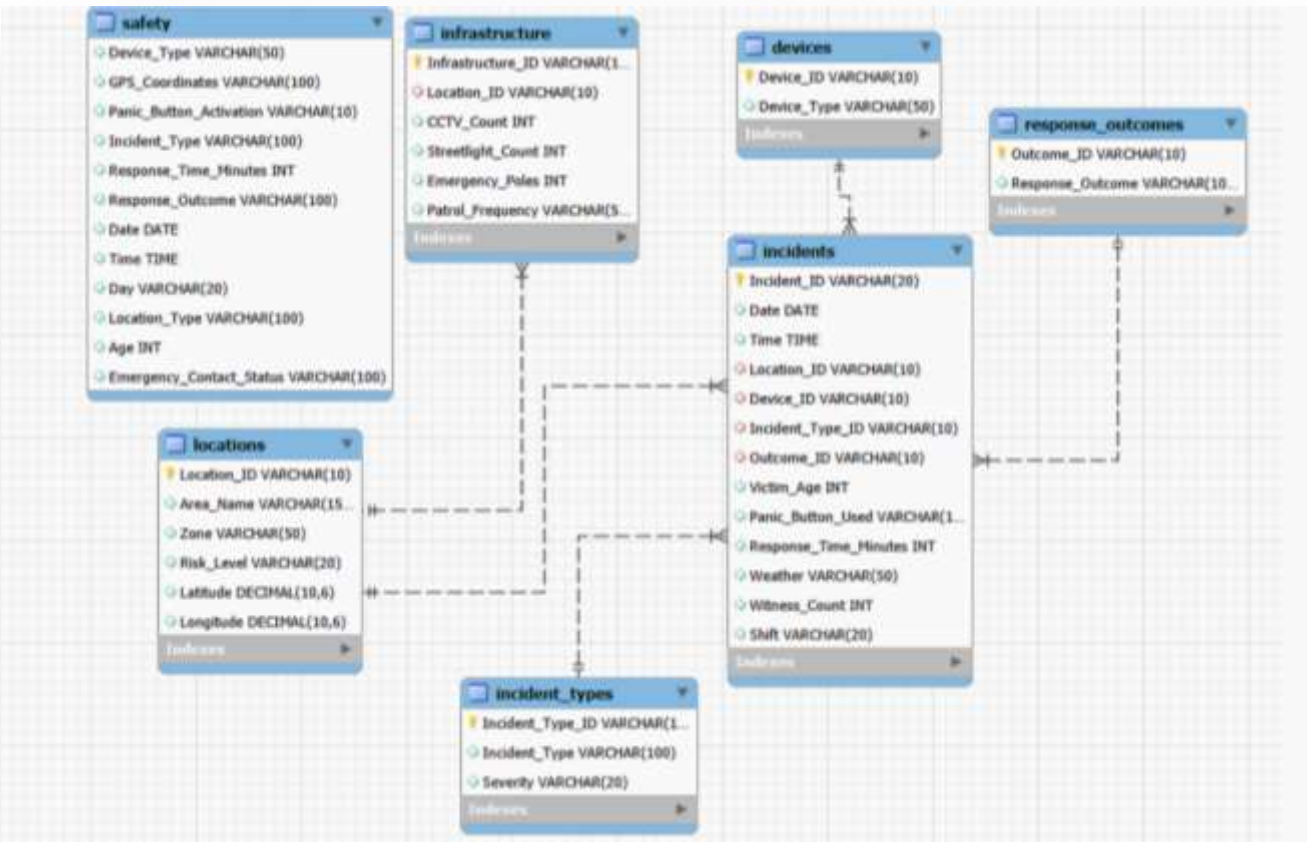


TABLE ROW COUNTS

incidents	20,000
safety (staging)	1,000
infrastructure	100
locations	100
incident_types	8
devices	3
response_outcomes	3

Data Quality & Validation

Before analysis, every table was checked for duplicates, nulls, and out-of-range values across all 20,000 incident records.



Data

Zero duplicate incident IDs and zero missing values across incident_id, date, time, location, outcome, age, response time, etc.
The dataset is ready for analysis without cleansing.

KPI Dashboard



20,000

TOTAL INCIDENTS



100

LOCATIONS AFFECTED



3

DEVICE TYPES IN USE



8

DISTINCT INCIDENT TYPES



56.65%

PANIC BUTTON ACTIVATIONS



38.92

AVERAGE VICTIM AGE (YRS)



12.37 min

AVERAGE RESPONSE TIME

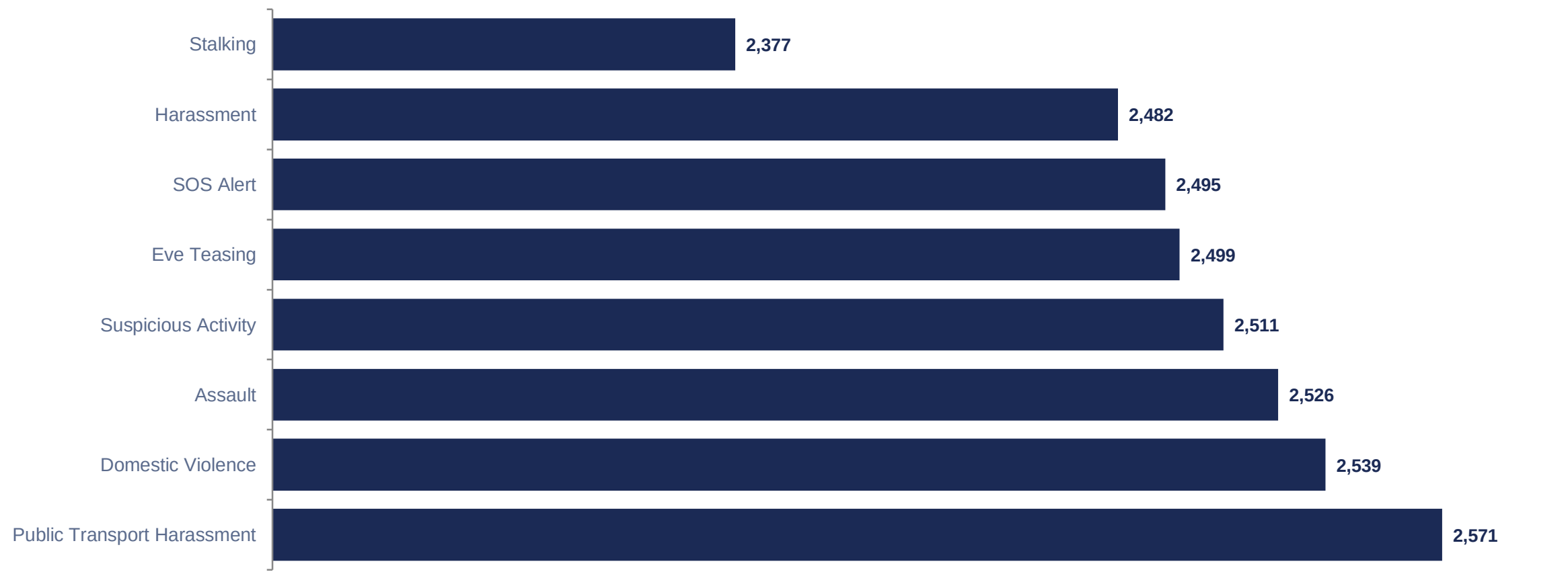


11,329

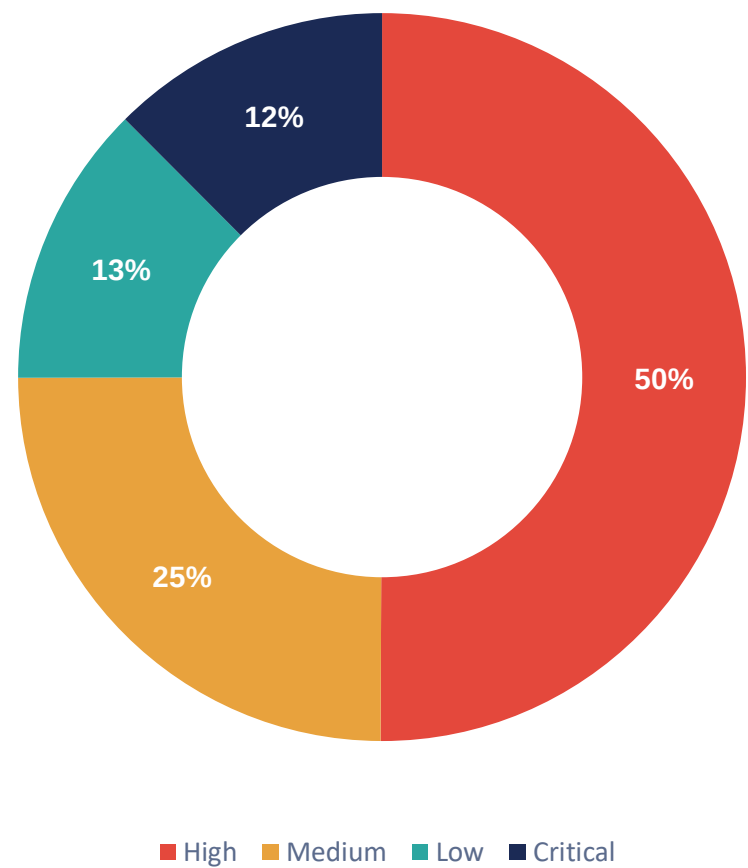
PANIC BUTTON USES (COUNT)

Incident Type Analysis

All eight incident types occur at strikingly similar volumes: no single category dominates, meaning safety interventions must be broad based rather than targeted at one issue.



Severity Distribution



Half of All Incidents Are High-Severity

Exactly 50.06% of incidents (10,013) are classified High severity — and Critical-severity ‘SOS Alert’ cases carry the same 12.4 minute average response time as every other category, suggesting response prioritization by severity is not yet happening operationally.

Severity	Count	% Total	Avg Resp.
High	10,013	50.06%	12.33 min
Medium	4,981	24.91%	12.40 min
Low	2,511	12.55%	12.47 min
Critical	2,495	12.47%	12.37 min

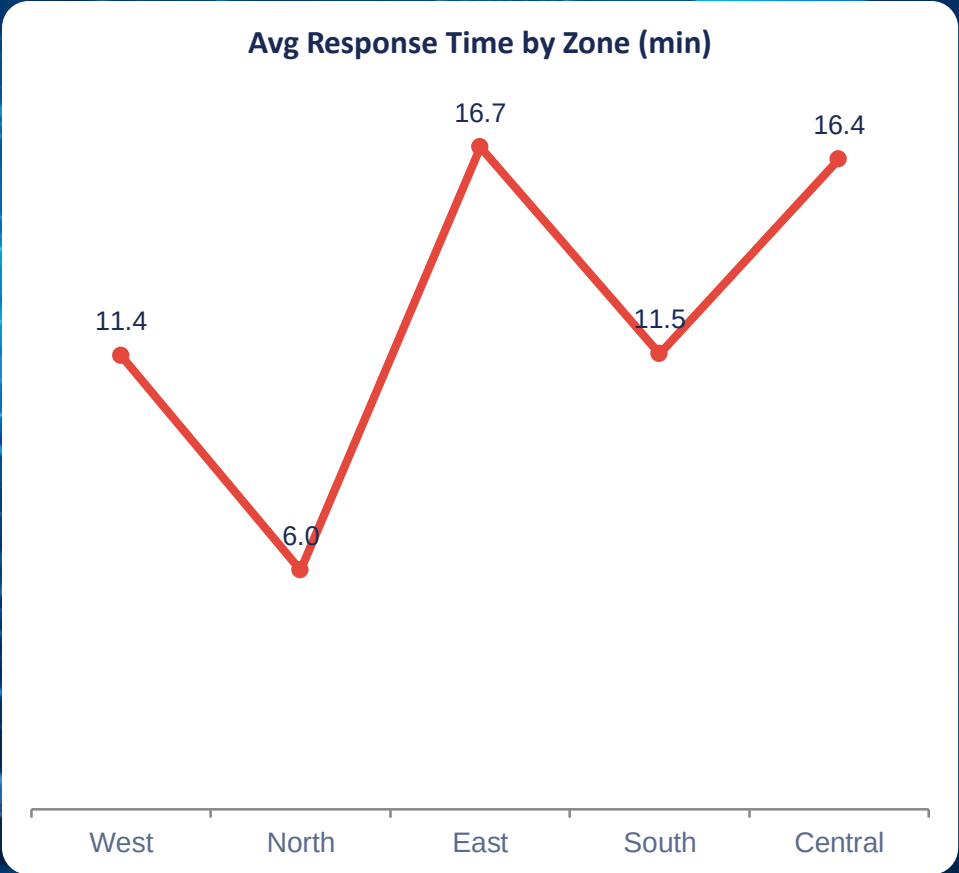
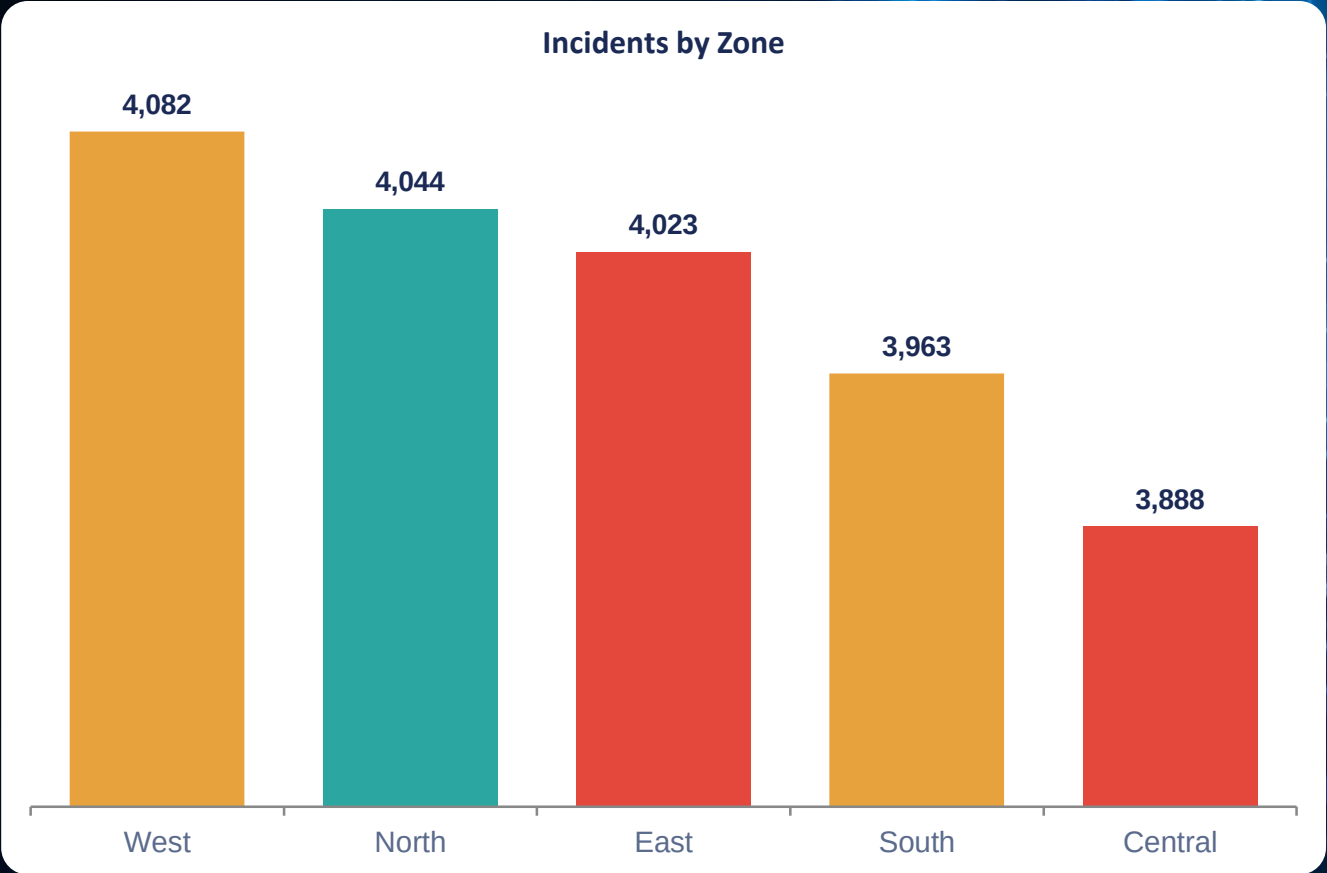
Top 10 Highest-Incident Locations

Ranked by total incident volume across all 100 tracked locations.

#	ID	Area Name	Zone	Risk	Incidents
1	L031	Savvy Subcity	East	High	235
2	L066	Technovia Heights	South	Medium	224
3	L044	Arama Residency	West	Medium	223
4	L069	Southlake Vista	South	Medium	222
5	L085	Silver Hub Gardens	North	Low	222
6	L062	Creme Delights Enclave	South	Medium	221
7	L042	Arama Extension	West	Medium	220
8	L013	Real Chartered Extension	Central	High	217
9	L099	Silver Hub Phase 1	North	Low	216
10	L014	K. Springs Phase 2	Central	High	216

Zone & Risk Level Patterns

City zones carry very different response burdens — East and Central (High risk) take nearly 3x longer to respond to than North (Low risk).



Response Time Analysis

2 min

FASTEST RESPONSE

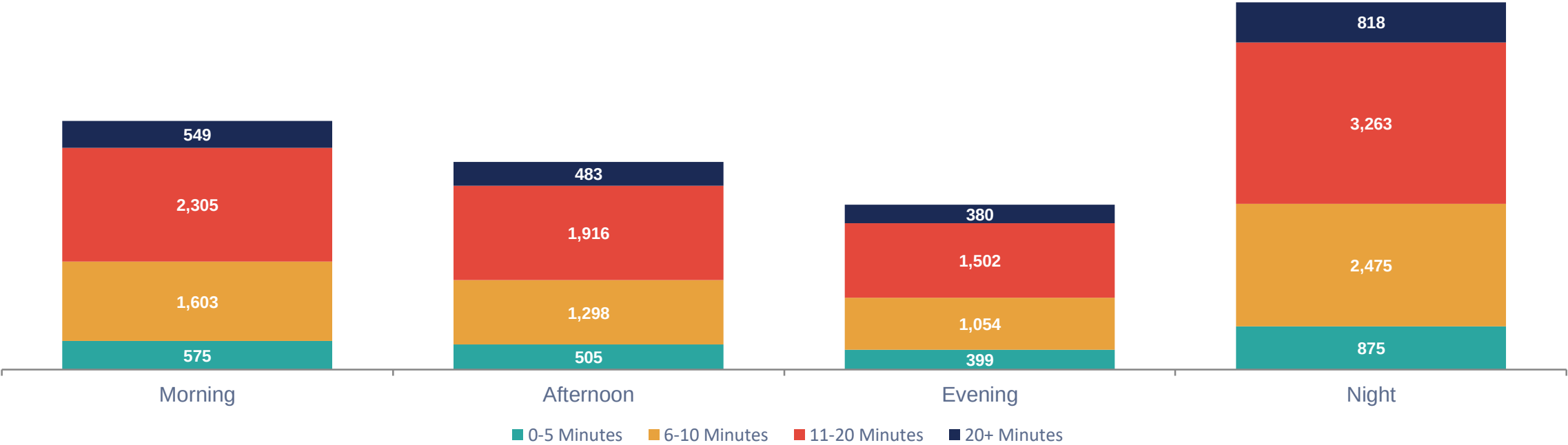
12.37 min

AVERAGE RESPONSE

25 min

SLOWEST RESPONSE

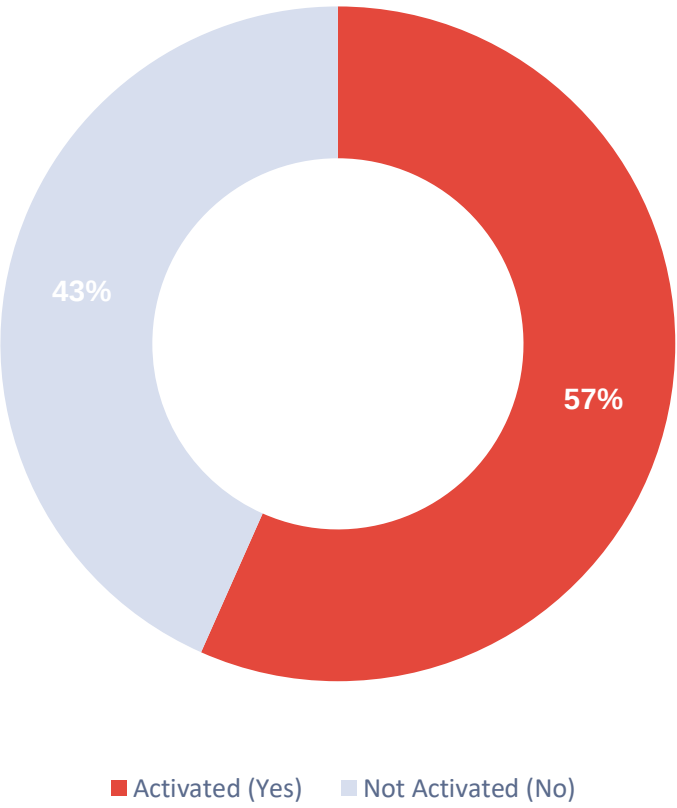
Response Category Volume by Shift



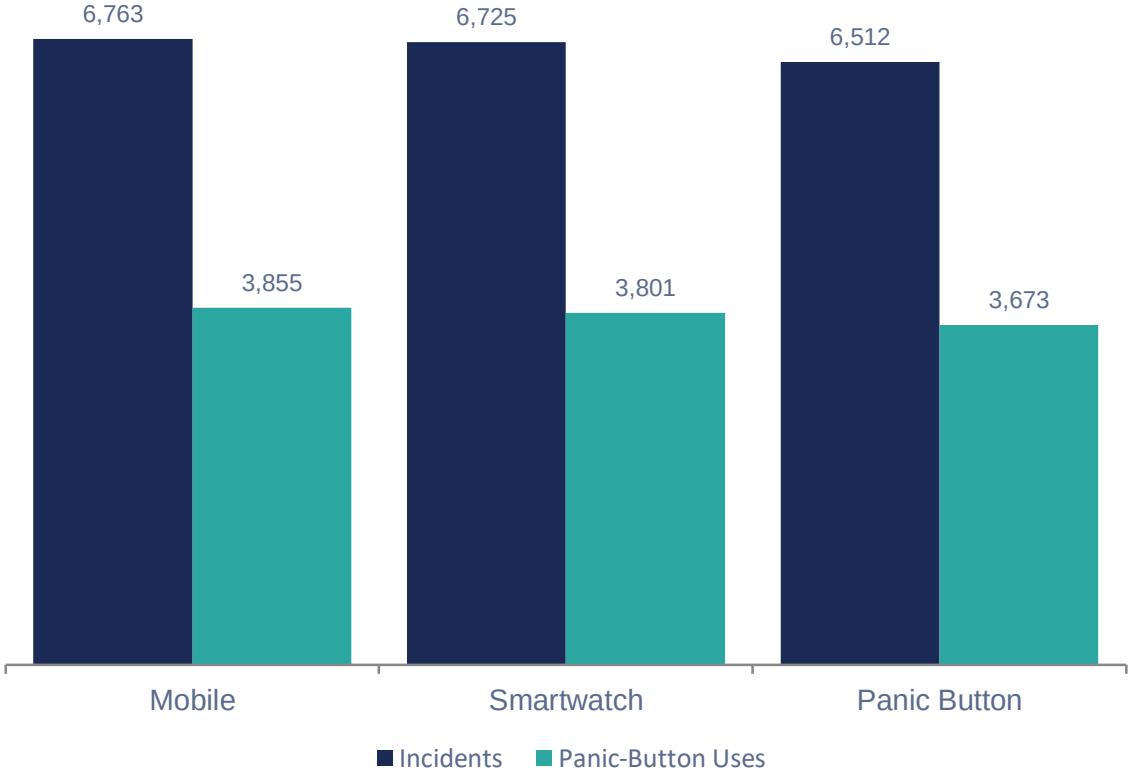
Panic Button & Device Usage



Panic Button Activation



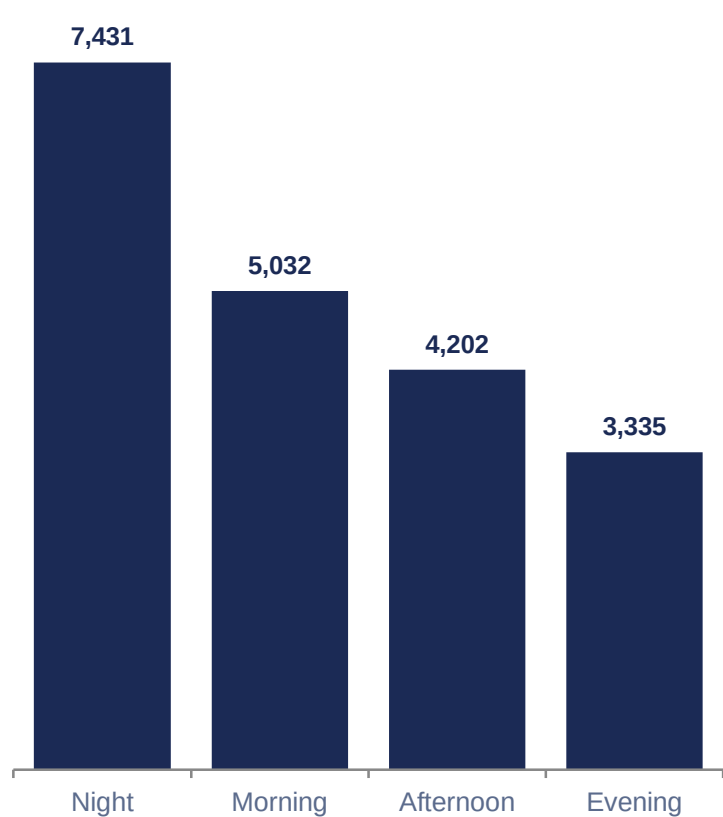
Incidents by Device Type



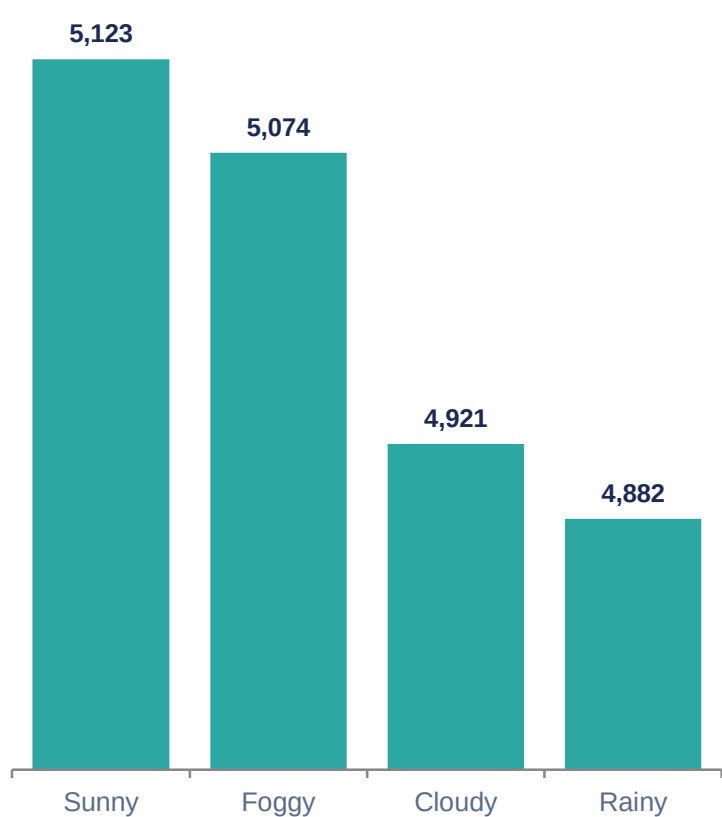
Shift, Weather & Witness Patterns

Night shift sees the most incidents (7,431) with the slowest average pace of activity; weather has little effect on volume (spread <5%); most incidents (50%+) occur with 3–5 witnesses present.

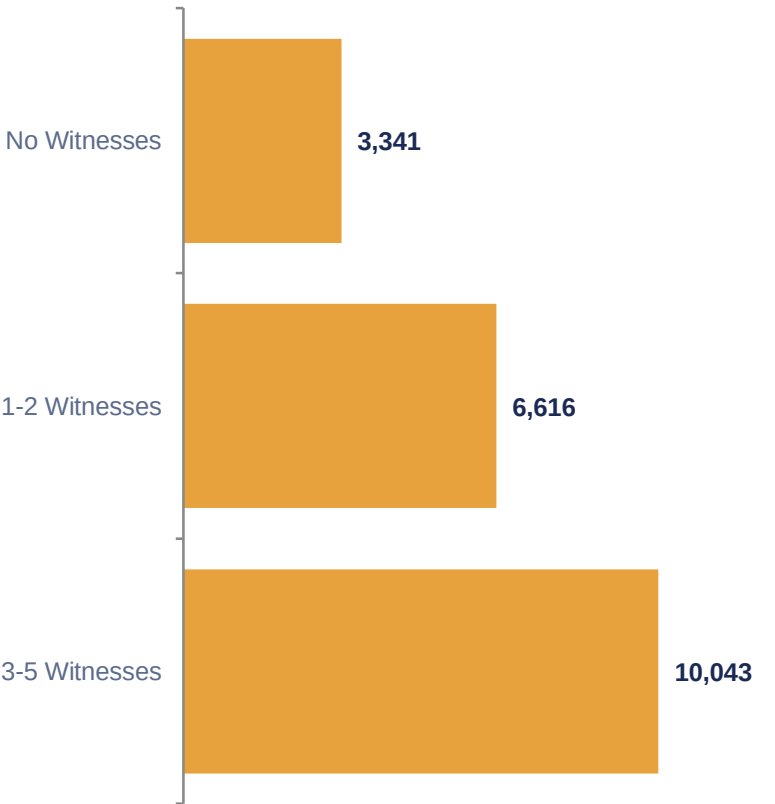
By Shift



By Weather



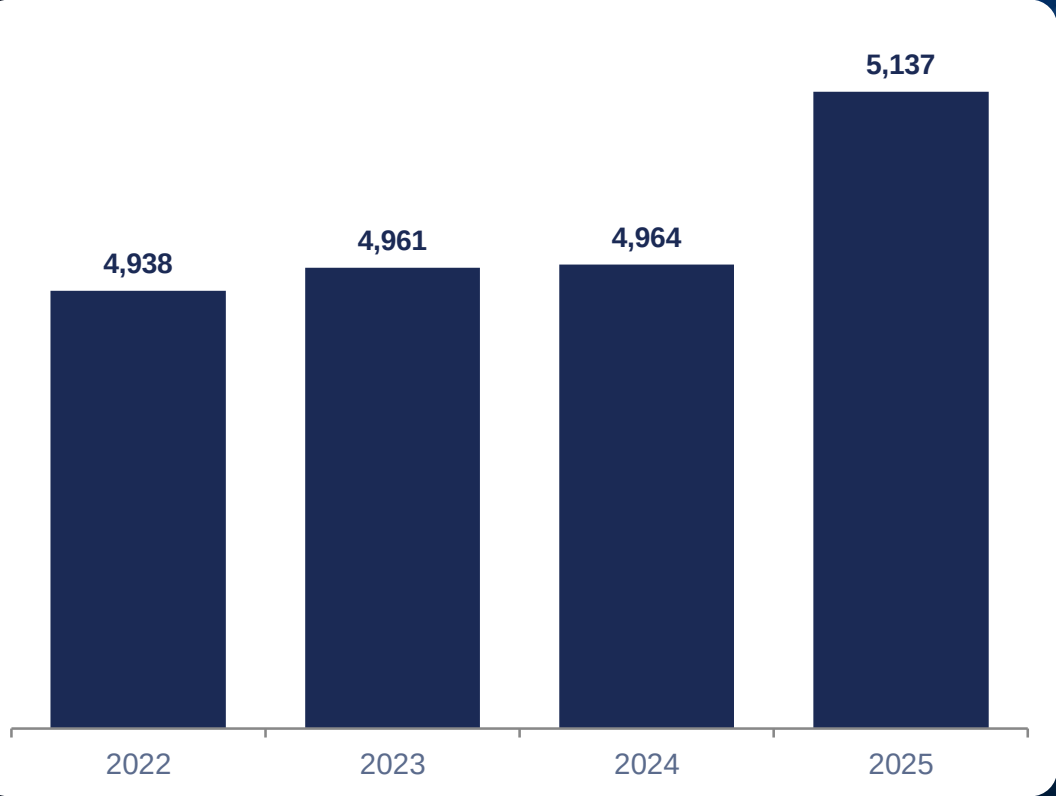
By Witness Count



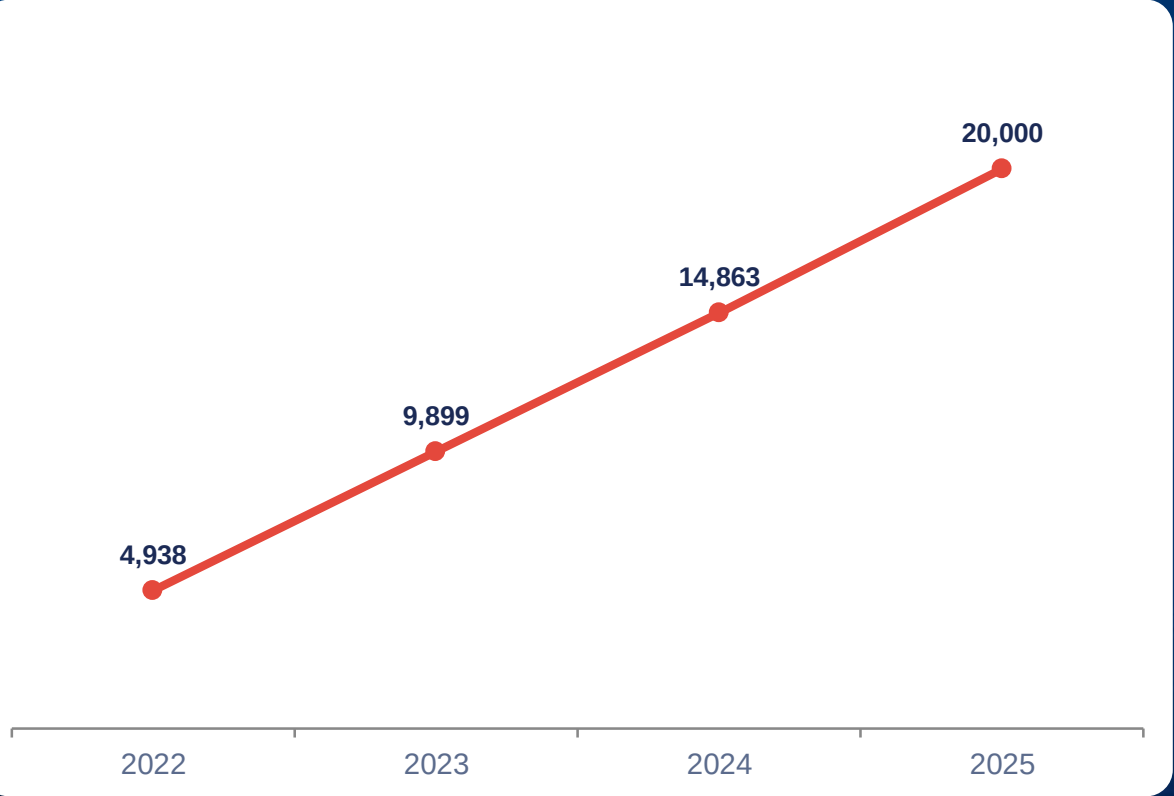
Incident Trends Over Time

Across 48 months of data (Jan 2022 – Dec 2025), yearly incident volume is remarkably stable — there is no runaway growth, but also no meaningful decline.

Annual Incident Volume



Cumulative Incident Growth



Safety Infrastructure Gaps

Cross-referencing incident load against CCTV, streetlighting, emergency poles and patrol frequency for the 5 highest-volume locations.

Area	Risk	CCTV	Lights	Poles	Incidents	Avg Resp.
Savvy Subcity	High	48	167	11	235	17.0 min
Technovia Heights	Medium	22	145	5	224	11.73 min
Arama Residency	Medium	28	144	5	223	11.43 min
Southlake Vista	Medium	38	140	5	222	11.40 min
Silver Hub Gardens	Low	5	54	3	222	5.92 min

Savvy Subcity has the most CCTV (48) and streetlights (167) of the top 5, yet the highest incident count and **slowest response time (17 min)**, hardware alone isn't solving the problem. Meanwhile Silver Hub Gardens, despite minimal infrastructure (5 CCTV), sees comparable incident volume but a far **faster response (5.92 min)**, pointing to patrol responsiveness as the deciding factor.

Business Priority Ranking Model

A composite score built with CTEs and DENSE_RANK() combines incident volume, response speed, and risk classification into one prioritization metric for city planners.

```
priority_score = incident_count + avg_response_time + (High risk: +20 · Medium: +10 · Low: +0)
```

- ❑ *Other advanced techniques applied across the 25-query suite: ROW_NUMBER() to find each location's top incident type, LAG() for month-over-month % change, and SUM() OVER() for running cumulative totals.*
- ❑ Window Functions: DENSE_RANK(), ROW_NUMBER()
- ❑ Common Table Expressions (CTEs): Modular WITH clauses for isolating severe incidents and building priority score models.
- ❑ Time-Series Analysis: Month over Month revenue growth velocity and ride frequency intervals via LAG() and DATEDIFF().

Key Insights

No dominant incident type

All 8 categories fall within a 200 incident band (2377 to 2571): the response strategy must cover the full spectrum, not one issue.

Half of incidents are High severity

50.06% of all 20,000 incidents are High severity, yet response time doesn't vary meaningfully by severity level (12.3 to 12.5 min across the board).

Response time isn't the bottleneck city wide

12.37 min average, ranging only 2 to 25 min: but East and Central zones (High risk) average 16–17 min while North's 6.02 min, a 3x gap.

Infrastructure volume ≠ safety outcome

Savvy Subcity has the most cameras & lights of any top location, yet the highest incident count and slowest response: patrol responsiveness matters more than hardware.

Panic button usage is moderate

56.65% of incidents involve panic-button activation; Mobile, Smartwatch and Panic Button devices are used almost equally (6500–6800 each).

Incident volume is flat, not falling

4938 → 5137 incidents per year (2022–2025), a slight uptick — no evidence current measures are reducing incident frequency.

Summary of Analytics Outcomes

Mission SafeCity examines women's safety incident records — collected via mobile apps, smartwatches, and public panic buttons — to help city safety planners understand where incidents cluster, how fast responders arrive, and which locations most urgently need infrastructure investment.



Validated Risk Patterns

Successfully mapped the correlation between time, location, and incident frequency across the City.



Flagged High-Risk Zones

Ranked locations and city zones by incident volume and risk level



Evaluated Response

Measured response speed across shifts, devices and panic button use



Infrastructure Impact

Proved the statistical value of functional CCTV and street lighting is deterring crimes.



Scored Priorities

Built a composite score to rank locations for intervention



Applied Advanced SQL

CTEs, window functions, ranking & lag analysis across 25+ queries

Thank You!

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[Github](#)